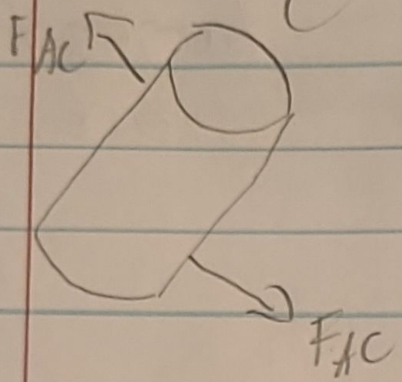
FBD:



$\tau = \frac{V}{A} \quad \frac{\text{Force}}{\text{cross section}}$

$\tau_{allowed} = \frac{\tau_y}{n} \rightarrow \tau = \frac{F_{max}}{A_{pin}} \le \tau_{allowed}$

$\frac{F_{max}}{A_{pin}} \le \frac{\tau_y}{n} \qquad A_{pin}\tau_y \ge F_{max}n$

$\tau_{allow} = \frac{170\text{ ksi}}{4} = 42.5\text{ ksi} \qquad A_{pin} = \frac{n F_{max}}{\tau_y}$

$F_{max} = 20 + \sqrt{(34.64\sin 60)^2 + (34.64\cos 60)^2} + \sqrt{(11.5417\cos 60)^2 + (11.5417\sin 60)^2}$

$F_{max} = 66.187$

$66.187\left(\frac{.224809\text{ kip}}{1\text{ kN}}\right) = 14.879\text{ kip}$

$A_{pin} = \frac{14.879}{42.5\text{ kip/in}^2} = .3501\text{ in}^2$

$A = \frac{\pi d^2}{4} \qquad d = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4(.3501)}{\pi}} = .667\text{ in}$

$V = A \cdot L \rightarrow W = \rho A L \cdot$ Number of pins

$W_{pins} = .278(.3501)(15.1575)(5)$

$W_{pins} = 7.376\text{ lbs}$